Layer Before Law

An Addressed Architecture for Relational Admissibility, Effective Law Selection, and Quantum-Gravitational Recovery

Mark Charles McLaughlin

ORCID: 0009-0005-7736-1511

Framework	McLaughlin-Kairos Unified Field Theory (MKUFT)
Publication status	Canonical public preprint - frozen standalone DOI release
Version	1.0
Canonical public date	17 July 2026
Revision	16 August 2026
Framework DOI	10.5281/zenodo.17780566
Standalone DOI	10.5281/zenodo.21971270

Version boundary. This PDF is the frozen standalone release of the canonical GitHub preprint. DOI 10.5281/zenodo.21971270 identifies this exact version upon publication; later revisions will not be silently backdated into this deposit.

> ***Version boundary. This is the frozen standalone release of the canonical GitHub preprint. DOI 10.5281/zenodo.21971270 identifies this exact version upon publication; later changes will remain later revisions rather than being silently backdated into this deposit.***

Abstract

Quantum theory and general relativity are extraordinarily successful within their demonstrated domains, yet their standard formalisms are not straightforwardly unified. *Layer Before Law* asks whether part of that difficulty may arise one conceptual step earlier: before selecting the physical law that governs a state, has the relevant object, context, admissibility structure, and scale been typed correctly?

Within the MKUFT research programme, the proposal is an addressed architecture rather than a completed theory of quantum gravity. A complete effective state is represented by a typed object carrying substrate participation, informational/relational structure, physical expression, observer-positioned registration, contextual address, and retained load-bearing history. From that addressed state, a compatibility map constructs an admissibility descriptor; a law assembler then constructs the active physical-law object - its admissible domain, transition rule, and any weighting or selection structure required by the physical model. Physical propagation occurs under that law object, measurement or registration produces the realised physical/record state, and the realised path readdresses the next effective possibility object.

The paper develops this architecture, adds a property-relative law-ownership gate and bidirectional scale readdressing, separates higher-law failure from constitutive closure and recoverability, applies the framework to entanglement as a worked constraint case, states explicit Bell/local-hidden-variable and no-signalling requirements, and defines recovery obligations for both quantum and gravitational regimes. It does **not** derive the Born rule, Bell correlations, quantum field theory, Einstein dynamics, or a fundamental quantum-gravity theory. Its present contribution is narrower: a typed, falsifiable research architecture for testing whether relational and contextual structure performs independent work beyond the strongest adequate physical-state/history/boundary description.

1. Problem statement

A common schematic target for quantum gravity is

$$\text{quantum theory} + \text{general relativity} \longrightarrow \text{deeper unified theory}.$$

That target is natural if the quantum and gravitational descriptions being combined are already expressed at the correct foundational level. *Layer Before Law* asks a prior question:

> ***Before asking which law governs an object, have we established the correct addressed object and the admissibility structure under which that law is physically applicable?***

This does not weaken the empirical authority of quantum theory or general relativity. It changes the order of the research task. The established theories become recovery targets that a deeper architecture must reproduce under declared regimes rather than ingredients whose foundational status is assumed in advance.

The central hypothesis is therefore not "quantum mechanics is wrong" or "general relativity is wrong." It is that a useful unification programme may need to distinguish:

1. what the effective object is;
2. which relations and boundary/context variables are load-bearing;
3. which physical states and paths are admissible;
4. which law object is active on that admissible domain;
5. how physical propagation and measurement occur; and
6. how the realised outcome changes the next addressed possibility object.

2. Literature boundary and novelty discipline

The possibility that spacetime or gravitational dynamics may be emergent is not unique to MKUFT. Jacobson derived the Einstein equation as an equation of state from local horizon thermodynamics [6]. AdS/CFT supplied a concrete duality between gravitational and non-gravitational descriptions [7]. Ryu and Takayanagi related boundary entanglement entropy to bulk geometry [8], and Van Raamsdonk argued that spacetime connectivity is intimately related to entanglement [9]. General relativity also has a controlled low-energy effective-field-theory treatment [10].

Likewise, non-separable quantum correlations are not an open empirical question in the simple local-hidden-variable sense. Bell's theorem [2], the CHSH formulation [3], and loophole-free Bell tests [4] sharply constrain local-realist reconstructions.

Accordingly, this paper does **not** claim originality for any of the following by themselves:
- emergent spacetime;
- relational or non-separable quantum structure;
- constraints and admissibility;
- effective field theory;
- measurement contextuality;
- state-space descriptions;
- the idea that macroscopic laws may be effective rather than fundamental.

The candidate MKUFT contribution is narrower: the explicit **address** -> **admissibility** -> **active-law-object** -> **propagation** -> **physical/record outcome** -> **readdressing** factorisation, together with typed layer discipline, a strongest-fair physical null, controlled-deformation tests, and explicit failure conditions.

The proposal therefore stands or falls on whether this particular organisation yields independent explanatory, mathematical, predictive, or interventional value.

3. Typed addressed architecture

MKUFT uses four active layer addresses:
- S: substrate or source-potential;
- I: informational and relational organisation, including identity, constraints, admissibility and rule-bearing structure;
- P: physical expression, including matter, fields, energy, geometry, timing, boundary conditions and measurable interaction;
- O: observer-positioned registration, including measurement context, record, perspective, and bounded observer-state participation where operationally defined.

These are **typed addresses**, not four ordinary spatial dimensions. Their recursive traversal is written

$$S \longrightarrow I \longrightarrow P \longrightarrow O \longrightarrow S.$$

The final return does not require that an observation rewrites an ambient universal substrate. Operationally, a realised physical/registered outcome changes the state, history, context and relations from which the next **effective possibility object** is computed.

The arrows are dependencies, not distances in a hidden geometry.

4. Complete effective state

Let the effective addressed state at time t be

$$\boxed{\mathcal{U}_t = \left(S_t^{\mathrm{eff}}, I_t, P_t, O_t, b_t, H_t\right)}$$

where
- S_t^{eff} is the effective substrate/possibility participation relevant to the current model;
- I_t is the typed informational-relational structure;
- P_t is the physical state;
- O_t is the registered/observer-positioned state where operationally required;
- b_t is extrinsic or contextual address; and
- H_t is the retained load-bearing history object.

Let E_t collect the measured or controlled environment required by the physical claim.

This is not a claim that all components live in one Euclidean vector space. It is a bookkeeping and typing object whose components may inhabit different mathematical categories.

5. Admissibility before active law

The first formal step is a typed compatibility or admissibility map

$$\boxed{\Xi_t = \mathcal{Q}_t(\mathcal{U}_t, E_t)}$$

where Ξ_t may contain support restrictions, boundary data, relation scopes, path/history constraints, symmetry requirements, compatibility classes, or other typed admissibility information. It need not be a scalar.

The active physical-law object is then

$$\boxed{\mathfrak{L}_{P,t} = (\mathcal{D}_{P,t}, \mathcal{T}_{P,t}, \mathcal{W}_{P,t})}$$

with

- $\mathcal{D}_{P,t}$: admissible physical state/path/operator domain;
- $\mathcal{T}_{P,t}$: the physical transition rule, generator, channel, or evolution law appropriate to the regime;
- $\mathcal{W}_{P,t}$: any regime-appropriate weighting, selection, or measure structure.

A typed law assembler gives

$$\boxed{\mathfrak{C}_P : (\Xi_t, P_t, E_t) \longmapsto \mathfrak{L}_{P,t}}$$

and the minimal *Layer Before Law* specialisation is domain conditioning,

$$\boxed{\mathcal{D}_{P,t} = \mathfrak{D}_P(\Xi_t)}$$

with the standard transition and weighting structure realised on that domain.

This minimal form is deliberately conservative. The general architecture permits Ξ_t to affect $\mathcal{D}$, $\mathcal{T}$, $\mathcal{W}$, or a typed combination **only where that stronger dependence is independently demonstrated**.

6. Propagation, measurement and recursive readdressing

The complete architectural update is

$$\boxed{\begin{aligned} \mathrm{Update}_{\mathrm{SIPO},\Delta}(\mathcal{U}_t) = \mathrm{Readdress}_{t+\Delta}\Big[\mathcal{U}_t,& \\ \mathrm{Instrument}_{P\to(P,O)}\Big(\mathrm{Propagate}^{\mathfrak{L}_{P,t}}_{P,\Delta}(P_t)\Big)\Big]&, \\ \mathfrak{L}_{P,t} = \mathfrak{C}_P(\mathcal{Q}_t(\mathcal{U}_t, E_t), P_t, E_t).& \end{aligned}}$$

with the readable composition

$$\boxed{\mathrm{Update}_{\mathrm{SIPO}} = \mathrm{Readdress} \circ \mathrm{Instrument}_{P\to(P,O)} \circ \mathrm{Propagate}^{\mathfrak{L}_P}_P, \qquad \mathfrak{L}_P = \mathfrak{C}_P \circ \mathcal{Q}}$$

subject to the typed state/context arguments shown above.

The measurement/registration instrument is typed as

$$\boxed{\mathrm{Instrument}_{P\to(P,O)} : P^-_{t+\Delta} \longmapsto (P_{t+\Delta}, O_{t+\Delta})}$$

so physical measurement back-action remains in P, while O carries the resulting record or registration where that distinction is useful.

The architecture can therefore be read as

```
complete addressed state + context/history
        ↓
typed admissibility descriptor
        ↓
active physical-law object (domain, transition, weighting)
        ↓
physical propagation
        ↓
physical measurement / registration
        ↓
realised path + record + context
        ↓
readdressed next effective possibility object
```

The staging is logical, not a claim that nature literally waits between discrete computational steps.

7. Layered address and physical individuality

For an object X in observational context C, define

$$\boxed{\mathcal{A}_X^{(C)} = (S_X, I_X, \mathcal{P}_X, O_C[X])}$$

where $\mathcal{P}_X$ denotes the complete physical address rather than a single spatial coordinate.

Two systems remain physically distinguishable when

$$\mathcal{P}_X \neq \mathcal{P}_Y.$$

The central layered-address principle is

$$\boxed{\mathcal{P}_X \neq \mathcal{P}_Y \quad \not\Rightarrow \quad I_X \text{ and } I_Y \text{ are completely separable}}$$

or, in plain language:

> ***distinct physical addresses do not require completely distinct informational relations.***

Within the broader MKUFT metaphysical hypothesis, a proposed substrate root may be written

$$r_S(S_X) = S_0,$$

but this is not used as a derived physical equation and does no empirical work in the entanglement argument unless a measurable consequence is independently specified.

7.1 Address before law ownership

The later MKUFT law-descent work sharpens the phrase *Layer Before Law* into a stricter burden: a real higher-order object does not automatically own a dynamically sufficient law for every property. The active law should be constructed at the coarsest address that has actually earned sufficiency for the property and intervention class being propagated.

Let

$$\pi_R : X \to Y$$

map lower-address states into a candidate higher relational address, and let $q : Y \to Q$ be the declared property of interest. For lower dynamics $F_X(x, u, e)$, property-relative law descent requires, to a declared tolerance,

$$\boxed{\pi_R(x) = \pi_R(x') \Longrightarrow q(\pi_R(F_X(x, u, e))) \approx q(\pi_R(F_X(x', u, e)))}$$

for the declared preparation, intervention class, environment, timescale and resolution. If the condition fails materially, the higher address may still describe a genuine organised whole, but it has not earned autonomous law ownership for q. The model must then retain the lower variables or move to a coupled multiscale address.

The lawful address motion is therefore bidirectional:

$$\boxed{\text{lower address} \rightleftarrows \text{effective / higher address} \rightleftarrows \text{coupled multiscale address}}$$

Lower-level physical laws are never erased by promotion. Promotion changes the address at which a declared downstream property is sufficiently represented; it does not delete the physical substrate that realises it.

This yields the stronger operational rule:

> ***Address before law ownership: construct the active law at the coarsest address that has demonstrated sufficiency for the property being propagated, and surrender that address when the sufficiency burden fails.***

7.2 Closure, recoverability and history

The current canon also separates three boundaries that the earlier Layer-Before-Law draft left too compressed:

1. loss of higher-address law sufficiency for a property q;
2. loss of the constitutive relation that makes the organised whole the relevant whole; and
3. loss of recoverability to a declared target relation under a declared control, environment and time horizon.

Let $M_D(q)$ denote a margin to loss of higher-address law sufficiency and M_C a margin to loss of constitutive closure. No universal ordering is assumed. A regime can therefore satisfy

$$\boxed{M_D(q) < 0, \qquad M_C > 0}$$

meaning that the simple higher-address law for q has failed while the organised whole remains intact.

For a target closure set $\mathcal{C}_R$, allowed intervention class U, environment class E, admissible domain $\mathcal{D}$ and horizon H, write

$$\mathrm{Rec}_H(x_t; \mathcal{C}_R, U, E) = 1$$

only when an admissible route exists that reaches the declared target class within the stated horizon. Recovery is therefore a reachability claim, not an assumption that degradation must reverse or that the return path is the time-reverse of the outward path.

Where transition and return thresholds differ,

$$\lambda_\uparrow \neq \lambda_\downarrow,$$

there is a candidate hysteresis signature. But apparent history dependence must first survive a missing-state test. If a bounded augmented state

$$Y_t^+ = \left(Y_t, H_t^{(k)}\right)$$

restores predictive closure, that history variable belongs in the address rather than being promoted as irreducible memory.

These distinctions matter directly to the SIPO update: readdressing is not merely the last arrow in a cycle. It is the mechanism by which the model changes the active explanatory scale when law sufficiency, closure, recoverability or required state changes.

8. Entanglement as a worked constraint case

The historical EPR problem [1] and Bell's theorem [2] make entanglement an unusually sharp test case because any proposed deeper architecture must reproduce non-separable correlations without collapsing into a forbidden local-hidden-variable model.

For physically distinguishable systems X and Y, write the candidate pair-level relational structure as

$$I_{XY} = (\tilde{I}_X, \tilde{I}_Y, R_{XY})$$

while maintaining

$$\mathcal{P}_X \neq \mathcal{P}_Y.$$

The proposal is therefore not physical co-location. It is the possibility of a pair-level relation R_{XY} that is not reducible to either endpoint alone.

Let x, y be local measurement settings and a, b the corresponding outcomes. Then

$$p(a, b \mid x, y, R_{XY})$$

denotes the joint outcome distribution.

A locally separable hidden-variable representation has the form [2,3]

$$\boxed{p(a, b \mid x, y) = \int d\lambda\, \rho(\lambda) p(a \mid x, \lambda) p(b \mid y, \lambda)}$$

and a viable Layer-Before-Law instantiation must **not** reduce R_{XY} to such an ordinary local instruction if it is intended to account for Bell-nonlocal correlations.

8.1 CHSH target

For binary outcomes $a, b \in \{-1, +1\}$, define

$$E(x, y) = \sum_{a,b} ab\, p(a, b \mid x, y)$$

and the CHSH combination [3]

$$\boxed{\mathcal{S} = E(0,0) + E(0,1) + E(1,0) - E(1,1)}$$

with local-hidden-variable bound

$$|\mathcal{S}| \leq 2.$$

Loophole-free experiments have observed violations of the local-realist bound [4]. Therefore an MKUFT branch that merely reconstructs a locally separable instruction set is excluded.

8.2 No-signalling target

Bell nonlocality is not a licence for controllable superluminal communication. A valid physical instantiation must preserve no-signalling [5]. The local marginal at Y must be independent of the distant setting x:

$$\boxed{\sum_a p(a, b \mid x, y, R_{XY}) = p_Y(b \mid y, R_{XY})}$$

and similarly

$$\boxed{\sum_b p(a, b \mid x, y, R_{XY}) = p_X(a \mid x, R_{XY})}$$

for all admissible distant settings.

The verbal statement "the relation belongs to the pair" is not enough. The no-signalling property must be derived from the instantiated dynamics.

8.3 Strongest-fair quantum null

The strongest fair null is that R_{XY} adds nothing beyond an adequate standard joint quantum state ρ_{XY} plus the complete physical preparation, measurement, boundary, history and environment description.

If replacing R_{XY} with that standard account leaves every prediction, intervention and explanation unchanged, then the Layer-Before-Law relational description is at most a translation layer for that case.

9. Quantum-gravitational recovery obligations

The architecture earns physical standing only by recovering established effective regimes.

Let $\Pi_{P,O}$ extract the relevant physical and registered output. For quantum-regime conditions $\mathcal{R}_Q$ and gravitational-regime conditions $\mathcal{R}_G$,

$$\boxed{\Pi_{P,O}[\mathrm{Update}_{\mathrm{SIPO}}(\mathcal{U};\mathcal{R}_Q)] \approx \mathcal{Q}_{\mathrm{eff}}}$$

and

$$\boxed{\Pi_{P,O}[\mathrm{Update}_{\mathrm{SIPO}}(\mathcal{U};\mathcal{R}_G)] \approx \mathcal{G}_{\mathrm{eff}}}$$

where every approximation requires a declared comparison object, statistic or norm, tolerance, uncertainty model and physical regime.

9.1 Quantum branch

A concrete quantum instantiation must identify whether the law object acts through Hilbert-space support, Hamiltonian/generator structure, channel structure, measurement instrument, weighting, or another declared physical object. It must then recover the standard quantum model for the controlled regime before any departure is considered.

Where density-matrix dynamics is used, admissible evolution must respect the positivity/normalisation structure required by the target formalism; where a closed-system unitary model is appropriate, the corresponding unitary limit must be recovered. No-signalling and Bell-compatible correlations remain non-negotiable constraints.

9.2 Gravitational branch

A gravitational instantiation must identify the admissible initial/boundary-data object, the active gravitational law object, and the recovery map to effective geometry.

In an appropriate low-energy classical regime, the target includes recovery of Einstein dynamics,

$$\boxed{G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}}$$

or an experimentally equivalent formulation, together with the associated consistency and conservation structure.

The existence of gravitational effective field theory [10] is an important null constraint: *Layer Before Law* cannot claim novelty merely because gravity admits low-energy quantum corrections or because an effective description exists.

10. What is actually being claimed

The present claim is architectural, not a completed fundamental law.

The paper contributes:

1. a typed separation between substrate participation, informational/relational structure, physical expression and observer-positioned registration;
2. a complete addressed-state object carrying contextual address and retained load-bearing history;
3. an explicit admissibility map $\mathcal{Q}$;
4. an active-law-object decomposition $(\mathcal{D}, \mathcal{T}, \mathcal{W})$;
5. a typed law assembler $\mathfrak{C}_P$;
6. a propagation/instrument/readdressing factorisation of the SIPO update;
7. explicit physical-only nulls;
8. Bell, no-signalling, quantum-limit and gravitational-limit recovery obligations;
9. a property-relative law-ownership gate that separates real higher-order organisation from demonstrated higher-address dynamical sufficiency;
10. bidirectional lower/higher/multiscale readdressing when that sufficiency changes;
11. separate law-sufficiency, closure and recoverability boundaries, with a state-augmentation guard before irreducible history is claimed;
12. controlled-deformation and falsification tests.

It does **not** presently claim:

- a derivation of quantum mechanics;
- a derivation of the Born rule;

- a derivation of Bell correlations;
- a derivation of quantum field theory;
- a derivation of Einstein's equations from MKUFT;
- a completed theory of quantum gravity;
- evidence that an independent $I \to P$ dynamics exists;
- that emergent spacetime or entanglement-geometry ideas are unique to MKUFT.

11. Controlled-deformation audit

The proposal is deliberately vulnerable to removal tests.

Remove the S-layer claim. If all physical predictions remain unchanged, universal substrate continuity is not empirically load-bearing in this paper.

Replace R_{XY} with the standard joint quantum state. If nothing explanatory, mathematical, predictive or interventional changes, the relational layer is a translation rather than a new mechanism for that case.

Remove the O layer. If registration does no distinct work, it should not be retained for symmetry alone. Physical measurement back-action remains a P-layer interaction.

Replace $\mathcal{Q}_t$ and $\mathfrak{C}_P$ with an adequate physical-only state/history/boundary/environment construction. If performance is unchanged, no independent informational-layer role has been established.

Flatten all typed spaces into one undifferentiated geometry. If nothing breaks, the layer architecture may be decorative. If the flattening creates invalid comparisons or shortcuts, the typing has at least methodological load.

Replace one SIPO architecture with unrelated effective laws. If the unrelated laws perform equally well without common explanatory loss, deeper unification has not been earned.

Force higher-address law ownership after its sufficiency test fails. If lower representatives mapped to the same higher address generate materially incompatible futures for the target property, the model must readdress downward or multiscale rather than preserving the simpler macro law by declaration.

Collapse loss of law sufficiency, closure and recoverability into one scalar. If the three boundaries separate under controlled perturbation, a one-variable account is falsified for that regime.

Claim irreducible hysteresis before state augmentation. If a bounded missing state or ordinary physical memory variable restores closure, that variable belongs in the address and the stronger irreducible-history claim fails.

12. Failure conditions

The proposal is weakened or rejected if any target branch materially requires one of the following failures:

1. S, I, P and O cannot be operationally separated where the claim requires them;
2. the proposed relational object adds no value beyond the strongest adequate physical description;
3. the entanglement account reduces to a Bell-incompatible local-hidden-variable instruction;
4. controllable superluminal signalling is permitted;
5. the instantiated update fails to recover the demonstrated quantum regime;
6. the gravitational branch fails to recover GR or an experimentally equivalent limit;
7. the meaning or type of a symbol changes silently between layers;
8. a formal object lacks a domain, physical referent, coupling, observable consequence or failure condition;
9. metaphysical interpretation is used to rescue a failed physical prediction;
10. no matched test can distinguish the MKUFT branch from a stronger existing model;
11. layer address, state-space dimension and physical spatial dimension are collapsed;
12. a claimed approximation lacks a comparison statistic, tolerance and regime;
13. the active law object cannot be operationally distinguished from an adequate physical-only state/history/boundary/environment model when independent $I \to P$ dynamics is claimed;
14. O is used to manufacture physical dynamics without a physical or independently tested coupling;
15. a higher relational address is granted law ownership for a property after representative, preparation, intervention, timescale or tolerance dependence has shown that the address is insufficient;
16. loss of property-specific law sufficiency is treated as loss of the organised whole without an independent closure test;
17. degradation is treated as irreversible collapse without a declared recoverability target and admissible route class;
18. apparent path dependence is called irreducible before bounded state/history augmentation has been tested.

13. Immediate research programme

The architecture is now sufficiently explicit that the next work should be branch-specific rather than adding another generic arrow.

Quantum programme

1. Choose the smallest system with mathematically exact and experimentally controllable domain/boundary structure.
2. Write the strongest standard physical-only model first.
3. Specify a candidate relational variable before looking at held-out outcomes.
4. State whether that variable changes support/domain, generator/channel, weighting, or only representation.
5. Recover the standard quantum limit and no-signalling constraints.
6. Test held-out/interventional performance against the physical-only null.
7. Promote the relational variable only if it supplies reproducible independent load.

Gravitational programme

1. Define the admissible initial/boundary-data object.
2. State the candidate addressed variable that is proposed to alter the law object.
3. Derive the effective geometric/dynamical output.
4. Recover Einstein/ADM behaviour in the declared regime.
5. Compare against GR/EFT and other strong alternatives.
6. State a prospective deviation or scaling relation before examining the decisive data.

Scale/address programme

1. Declare the lower and candidate higher addresses before outcome inspection.
2. Name the target property q and the intervention class for which higher-address sufficiency is claimed.
3. Test representative consistency or its stochastic/approximate analogue at a preregistered tolerance.
4. Track loss of law sufficiency $M_D(q)$ separately from constitutive closure M_C.
5. Define a target recovery set and test Rec_H rather than inferring reversibility from appearance.
6. Test bounded state/history augmentation before declaring irreducible hysteresis.
7. Readdress downward, upward or multiscale when the demonstrated sufficiency changes.

14. Conclusion

Quantum theory and general relativity do not need to be diminished for their foundational relationship to be questioned. *Layer Before Law* proposes that a unification attempt may sometimes begin with the wrong operation: selecting or combining effective laws before establishing the complete addressed object and its admissibility structure.

The resulting research architecture is

$$\boxed{\begin{array}{c}\text{address} \longrightarrow \text{admissibility} \longrightarrow \text{active law object} \longrightarrow \text{physical propagation} \longrightarrow \\ \text{measurement/registration} \longrightarrow \text{readdressing}\end{array}}$$

The architecture is now more than the unnamed update arrow of the earliest Layer-Before-Law draft. It has a typed state, a compatibility descriptor, an explicit physical-law object, a propagation/instrument chain, physical-only nulls, concrete recovery obligations, and an evidence-triggered rule for **which address may own the active law for a declared property**. It also keeps loss of law sufficiency, loss of constitutive closure and loss of recoverability as distinct boundaries rather than collapsing them into one vague notion of failure.

What remains is the part that matters most scientifically: instantiate the architecture in controlled physical regimes and determine whether it predicts or explains anything that the strongest existing physical account does not.

That is the line between an attractive architecture and a physical theory, and this paper leaves that line visible on purpose.

References

1. Einstein, A., Podolsky, B., & Rosen, N. (1935). "Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?" *Physical Review*, **47**, 777-780. DOI: 10.1103/PhysRev.47.777.
2. Bell, J. S. (1964). "On the Einstein Podolsky Rosen Paradox." *Physics Physique Fizika*, **1**, 195-200. DOI: 10.1103/PhysicsPhysique-Fizika.1.195.
3. Clauser, J. F., Horne, M. A., Shimony, A., & Holt, R. A. (1969). "Proposed Experiment to Test Local Hidden-Variable Theories." *Physical Review Letters*, **23**, 880-884. DOI: 10.1103/PhysRevLett.23.880.

4. Hensen, B., Bernien, H., Dreau, A. E., et al. (2015). "Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres." *Nature*, **526**, 682-686. DOI: 10.1038/nature15759.
5. Simon, C., Buzek, V., & Gisin, N. (2001). "No-Signaling Condition and Quantum Dynamics." *Physical Review Letters*, **87**, 170405. DOI: 10.1103/PhysRevLett.87.170405.
6. Jacobson, T. (1995). "Thermodynamics of Spacetime: The Einstein Equation of State." *Physical Review Letters*, **75**, 1260-1263. DOI: 10.1103/PhysRevLett.75.1260.
7. Maldacena, J. M. (1998). "The Large N Limit of Superconformal Field Theories and Supergravity." *Advances in Theoretical and Mathematical Physics*, **2**, 231-252. DOI: 10.4310/ATMP.1998.v2.n2.a1; arXiv: `hep-th/9711200`.
8. Ryu, S., & Takayanagi, T. (2006). "Holographic Derivation of Entanglement Entropy from the anti-de Sitter Space/Conformal Field Theory Correspondence." *Physical Review Letters*, **96**, 181602. DOI: 10.1103/PhysRevLett.96.181602.
9. Van Raamsdonk, M. (2010). "Building up spacetime with quantum entanglement." *General Relativity and Gravitation*, **42**, 2323-2329. DOI: 10.1007/s10714-010-1034-0.
10. Donoghue, J. F. (1994). "General relativity as an effective field theory: The leading quantum corrections." *Physical Review D*, **50**, 3874-3888. DOI: 10.1103/PhysRevD.50.3874.
11. McLaughlin, M. C. *Unified Field Theory. McLaughlin-Kairos. MKUFT*. DOI: 10.5281/zenodo.17780566.
12. McLaughlin, M. C. (2026). *Active Traversal and Load-Bearing Dependency in Typed Knowledge Architectures: A Matched-Control Evaluation Protocol for AI Systems*. Version DOI: 10.5281/zenodo.21341521.

Appendix A. Public source provenance

This standalone paper is assembled from the following public MKUFT canon modules. Paths identify the versioned source tree in the public MKUFT repository; they are provenance routes, not additional claims.

- `docs/01_MKUFT_CORE_EXTENDED.md`
- `docs/02_MKUFT_MATH_APPENDIX.md`
- `docs/13_PHYSICS_FACING_MKUFT_EXPLANATION.md`
- `docs/22_CROSS_LAYER_INVARIANTS_AND_LAYER_ADDRESSING.md`
- `docs/24A_ACTIVE_TRAVERSAL_AND_FUNCTIONAL_EMERGENCE_HYPOTHESIS.md`
- `docs/24B_STRONGEST_FAIR_NULL_AND_RELATIONAL_SPECIFICITY.md`
- `docs/25_LOAD_BEARING_INVARIANTS_AND_WHOLE_SYSTEM_DEFORMATION.md`
- `docs/27_TYPED_TRAVERSAL_AND_EQUATION_HYGIENE.md`
- `docs/32_RECURSIVE_CONSTRAINT_CLOSURE_AND_REACHABLE_STATE_GEOMETRY.md`
- `docs/32S3_RELATIONAL_BRACKETS_COMPLETION_GEOMETRY_AND_I_TO_P_ADMISSIBILITY.md`
- `docs/32S4_INTRINSIC_EXTRINSIC_ADDRESS_TRANSPORT_HOLONOMY_AND_BOUNDARY_CONDITIONED_REALISATION.md`
- `docs/33_SIPO_CAPSTONE_CONSTRAINT_CONDITIONED_ADDRESSED_UPDATE_LAW.md`
- `docs/33S1_DYNAMIC_INTERFACE_PROMOTION_AND_RECURSIVE_BOUNDARY_CLOSURE.md`
- `docs/33S2_RELATIONAL_CLOSURE_LAW_DESCENT_AND_BIDIRECTIONAL_READDRESSING.md`
- `docs/33S3_CROSS_SCALE_PERFORMANCE_RECOVERABILITY_AND_HYSTERETIC_READDRESSING.md`
- `docs/33A_LAW_DESCENT_AND_RECOVERABILITY_NOVELTY_AUDIT.md`